

Differential Equations in Geophysical Fluid Dynamics

XXVII. Coastal upwelling using reduced gravity model

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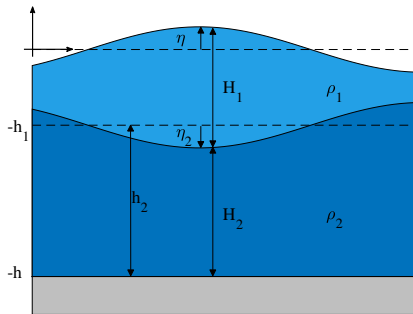
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Recap

Averaging over each density layer yields the governing equations of the layered ocean model:



Recap

≈ 1 (Boussinesq approximation)

$$\frac{\partial \bar{u}^{(1)}}{\partial t} - f \bar{v}^{(1)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial x} \quad (1a)$$

$$\frac{\partial \bar{u}^{(2)}}{\partial t} - f \bar{v}^{(2)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial x} - g' \frac{\partial \eta_2}{\partial x} \quad (2a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f \bar{u}^{(1)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial y} \quad (1b)$$

$$\frac{\partial \bar{v}^{(2)}}{\partial t} + f \bar{u}^{(2)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial y} - g' \frac{\partial \eta_2}{\partial y} \quad (2b)$$

$$\frac{\partial \eta}{\partial t} + h_1 \left(\frac{\partial \bar{u}^{(1)}}{\partial x} + \frac{\partial \bar{v}^{(1)}}{\partial y} \right) - \frac{\partial \eta_2}{\partial t} = 0 \quad (1c)$$

$$\frac{\partial \eta_2}{\partial t} + h_2 \left(\frac{\partial \bar{u}^{(2)}}{\partial x} + \frac{\partial \bar{v}^{(2)}}{\partial y} \right) = 0 \quad (2c)$$

where $g' = g(\rho_2 - \rho_1)/\rho_0$ is the reduced gravity.

Recap

Once **motionless bottom layer is assumed**, it becomes reduced gravity model:

$$\frac{\partial \bar{u}^{(1)}}{\partial t} - f \bar{v}^{(1)} = -g' \frac{\partial H_1}{\partial x} \quad (3a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f \bar{u}^{(1)} = -g' \frac{\partial H_1}{\partial y} \quad (3b)$$

$$\frac{\partial H_1}{\partial t} + \frac{\partial (H_1 \bar{u}^{(1)})}{\partial x} + \frac{\partial (H_1 \bar{v}^{(1)})}{\partial y} = 0 \quad (3c)$$

Coastal upwelling using reduced gravity model

Let us consider approximated reduced gravity model:

Anisotropic ocean

$$\frac{\partial \bar{u}^{(1)}}{\partial t} - f \bar{v}^{(1)} = -g' \frac{\partial H_1}{\partial x} \quad (4a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f \bar{u}^{(1)} = -g' \frac{\partial H_1}{\partial y} + \frac{\tau_y^s(t)}{\rho_0 h_1} \quad (4b)$$

$$\frac{\partial H_1}{\partial t} + \boxed{h_1} \frac{\partial \bar{u}^{(1)}}{\partial x} + \frac{\partial (H_1 \bar{v}^{(1)})}{\partial y} = 0 \quad (4c)$$

linearity ($H_1 \approx h_1$)

No alongshore gradient
($\partial/\partial y \approx 0$)

1. When is the no alongshore gradient (1D) assumption valid?
2. What is an anisotropic ocean?

Anisotropic ocean assumption

Consider large-scale ($Ro \ll 1$) coastal ocean having long alongshore scale and narrow cross-shore scale.

So cross-shore variables (x and $\bar{u}^{(1)}$) and alongshore variables (y and $\bar{v}^{(1)}$) have different scale¹:

$$x = L_x x^* \quad (5a) \qquad y = L_y y^* \quad (5c)$$

$$\bar{u} = U \bar{u}^* \quad (5b) \qquad \bar{v} = V \bar{v}^* \quad (5d)$$

For simplicity, let us consider $U \approx L_x/T$ and $V \approx L_y/T$, so $L_x \ll L_y$ alludes $U \ll V$. On the other word, $L_x/L_y \approx U/V \equiv \epsilon_{xy}$

$$\left(\underbrace{\frac{1}{fT}}_{Ro: \text{ small}} \underbrace{\frac{U}{V}}_{\epsilon_{xy}: \text{ small}} \right) \frac{\partial \bar{u}^*}{\partial t^*} - \bar{v}^* = -\frac{\partial H^*}{\partial x^*} \qquad \left(\frac{1}{fT} \underbrace{\frac{V}{U}}_{1/\epsilon_{xy}: \text{ big!}} \right) \frac{\partial \bar{v}^*}{\partial t^*} + \bar{u}^* = -\frac{\partial H^*}{\partial y^*}$$

$Ro: \text{ small}$ $Ro: \text{ small}$

¹Hereafter, dropping layer index.

Coastal upwelling using reduced gravity model

Solving (4) with respect to H yields

$$\frac{g'h_1}{f^2} \frac{\partial^2 H}{\partial x^2} - H = -h_1 \quad (6)$$

With initial condition $H|_{t=0} = h_1$

Closed boundary condition ($\bar{u}|_{x=0} = 0$) with (4) yields

$$\left. \frac{g'}{f} \frac{\partial H}{\partial x} \right|_{x=0} = \int_0^t \frac{\tau_y^s}{\rho_0 h_1} dt \quad (7)$$

And consider unperturbed offshore ocean:

$$\lim_{x \rightarrow \infty} H = h_1 \quad (8)$$

Coastal upwelling using reduced gravity model

Solution to the problem is given by

$$H = h_1 - \left(\frac{1}{Rd'} \int_0^t \frac{\tau_y^s}{\rho_0 f} dt \right) e^{-x/Rd'} \quad (9)$$

where $Rd' = \sqrt{g'h_1}/f$ is the internal Rossby radius of deformation.

Cross-shore length scale of the coastal upwelling is the Rossby radius of deformation!

Role of anisotropic ocean assumption

Cushman-Roisin and Beckers (2011) discussed

No anisotropic ocean assumption!

↓

$$\frac{\partial \bar{u}^{(1)}}{\partial t} - f \bar{v}^{(1)} = -g' \frac{\partial H_1}{\partial x} \quad (10a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f \bar{u}^{(1)} = \frac{\tau_0^s}{\rho_0 h_1} \sin(w_0 t) \quad (10b)$$

$$\frac{\partial H_1}{\partial t} + h_1 \frac{\partial \bar{u}^{(1)}}{\partial x} = 0 \quad (10c)$$

that explicitly show role of the anisotropic ocean assumption and explain why it also referred to as “long-wave approximation” (Brink, 2023).